Section 8 Signal cables

Contents

1. GENERAL

Connect all signal cables according to the drawings:

Z5 – 2 1	Signal cable plan magnet
Z5 – 2.2	Patient support MT
Z5 – 2.3.1	Gradient parts
Z5 – 2.4	RF parts
Z5 – 2.5	Console

Due to the fact that there is limited space between magnet covers and magnet, cables must be mounted with care. If you do not install the cables "in a nice manner" as indicated in this section, you will face problems installing the magnet covers.

Mount the optical fibres after all other cables have been installed (see chapter "Mounting the optical fibres").

CAUTION

Optical fibres are very delicate. Be careful when working with optic fibres.

NOTE

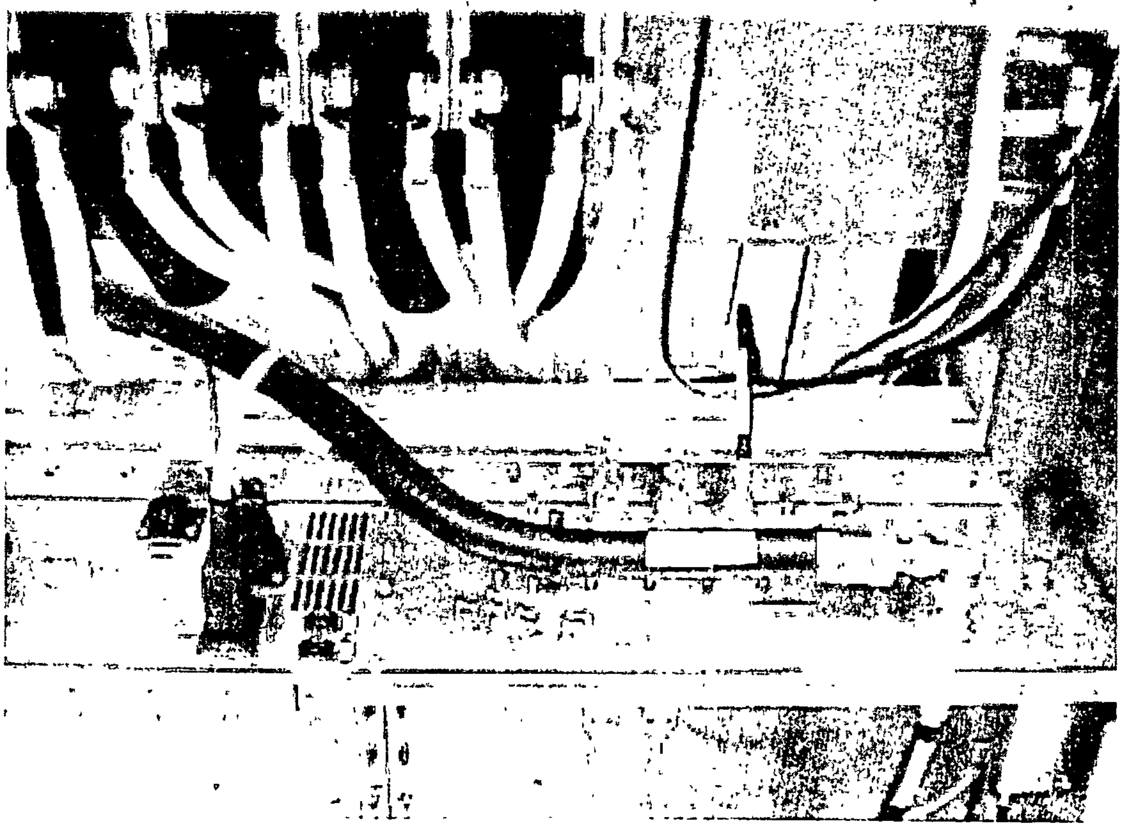
Cables that are not connected and cables that can move against other cables can cause electrical discharge (spikes) with image artifacts as a result. Root cables properly in ducts and in the hooks, present around the magnet.

2. INSTALLATION OF THE RF HIGH POWER CABLE

This paragraph is only valid for Intera 1.0 T systems

Intera 1.0 T system with Standard and Power gradients uses a thick 25 kW RF power cable.

Figure 1 - RF high power cable



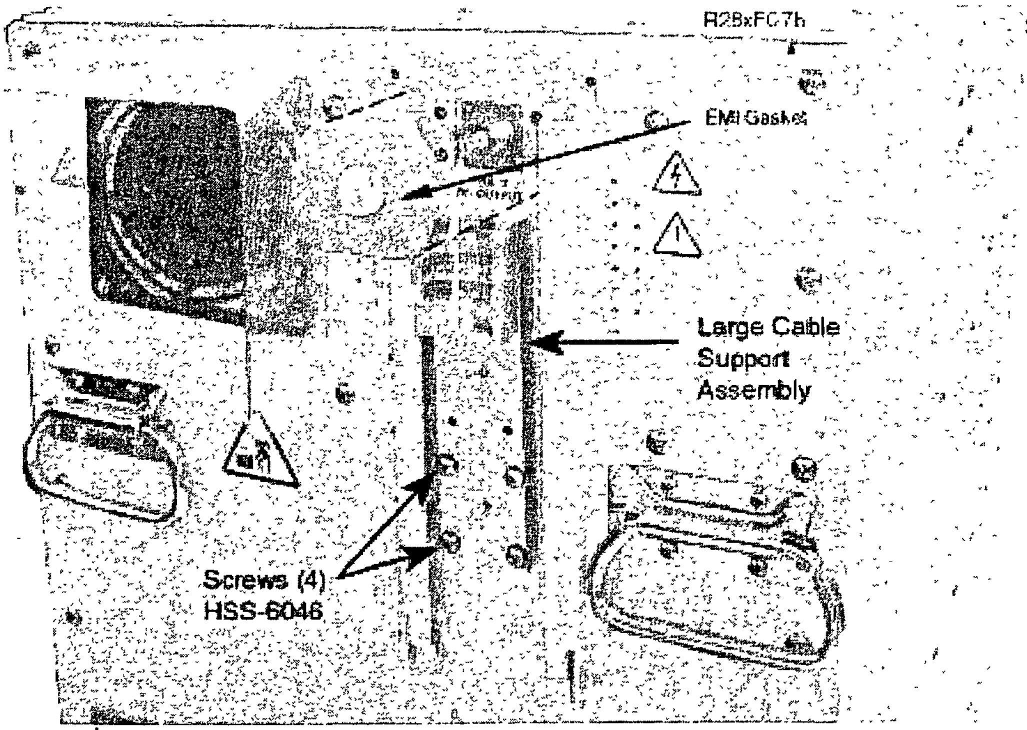
The high power cable between power amplifier and system filter box is needed in case 15 kW RF (Intera 1.5 T) power is used. Note that the minimum bending radius of this stiff cable is 300 mm. The power cable has connectors on both sides. You may want to shorten the power cable. For that purpose a "service set heliax cable" is available that you can order with the 12 NC number 4522 131 54431. A connector, tools and instructions are included in this set.

In the examination room a thin RF cable type RG217 is used.

Use the delivered L piece at the RF power amplifier output connector.

If you have an S23/S24 amplifier the RF-Output is on the backside of the amplifier.

Figure 2 - showing RF output connector and cable support assembly

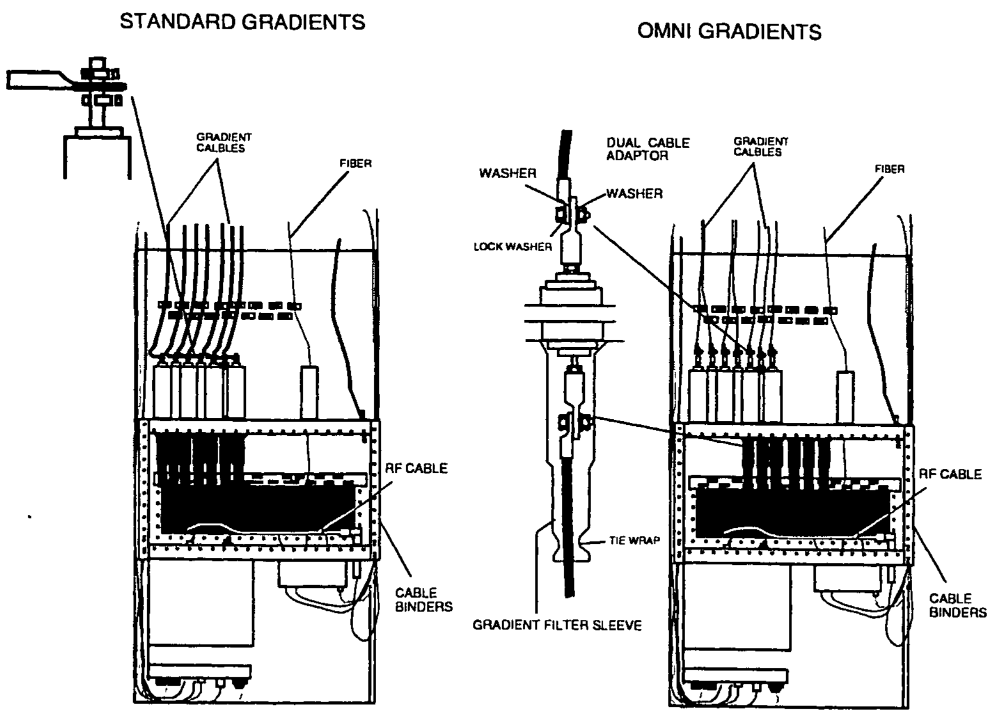


3. SYSTEM FILTER BOX CABLING

Mount the Gradient filters:

The gradient filters ZF170 to ZF177 are factory mounted.
Note that the gradient cables for the Omni gradients are connected differently to the filters compared with the Intera Standard situation. The gradient cable ends, which has black sleeves (holding two gradient cables together) is mounted to the gradient coil, and the other end is mounted to the system filter box.

Figure 3 - Cabling S.F.B.



Only for the Omni gradients, gradient filter sleeves are needed to prevent service engineers from touching the gradient connections by sticking their hand inside the system filter box from the outside of the examination (RF) room.

1. Slide the gradient filter sleeves over the gradient cables that leave the examination room (6x).
2. Connect the gradient cables to the dual cable adaptors.
3. Slide the gradient filters sleeves all the way up and secure them with a tie-wrap (6x).

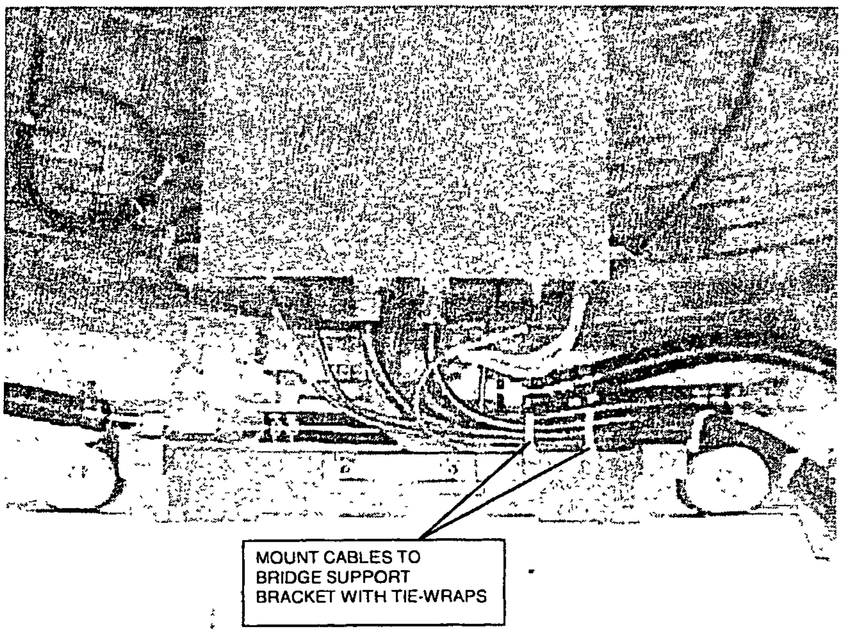
CAUTION

Take special care not to damage the gradient filters inside the system filter box. They are made of ceramic material and cannot withstand large forces; use two spanners to tighten the cable connection and use them in such a way that no force is applied to the filter.

4. HYBRID BOX / QBC CABLING

Cables from the hybrid box must always be routed via the right side of the magnet going up. Attache these cables with tie-wraps to the bradge support bracket.

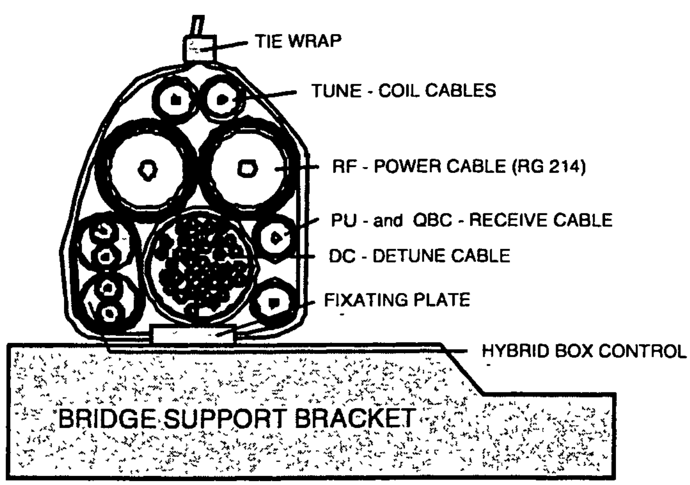
Figure 4 - Hybrid box cabling



The Hybrid box / QBC cables must always remain bundled as shown in the next figure.

Figure 5 – QBC/Hybrid cables

The QBC/Hybrid cables are bundeled



5. CABLE ROUTING AT MAGNET FRONT WITH PICU MOUNTERD AT RIGHT SIDE

Cable routing in this area is very critical. Be sure that no cable crossings are made. If you cross cables, magnet covers will not fit.

Figure 6 – Cable routing front (right side)

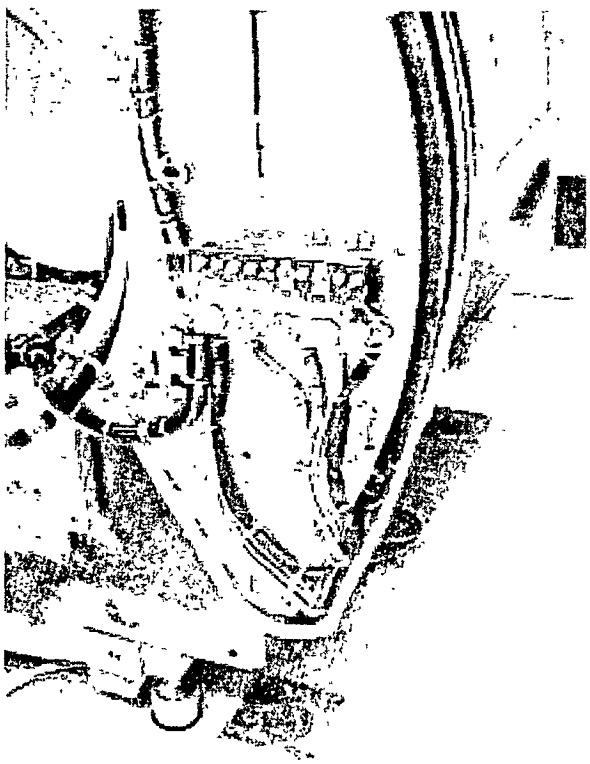
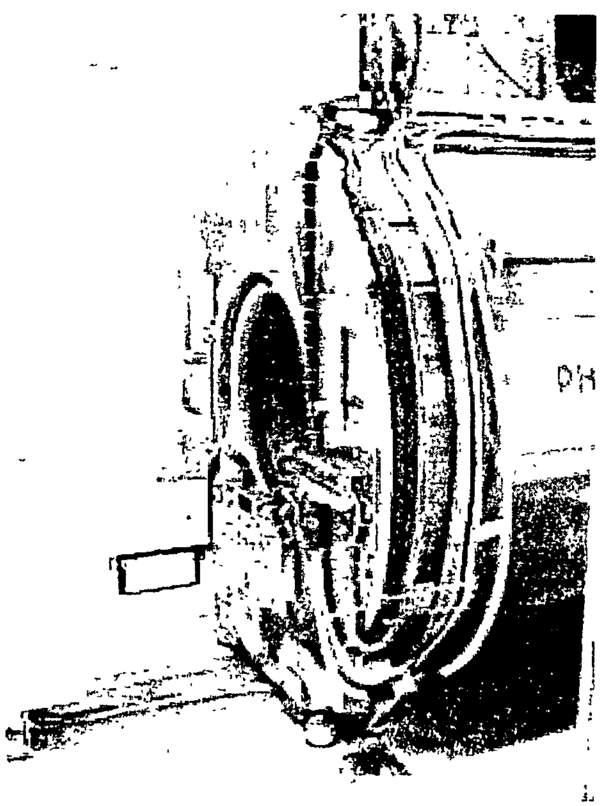


Figure 7 – Cable routing front (right side)

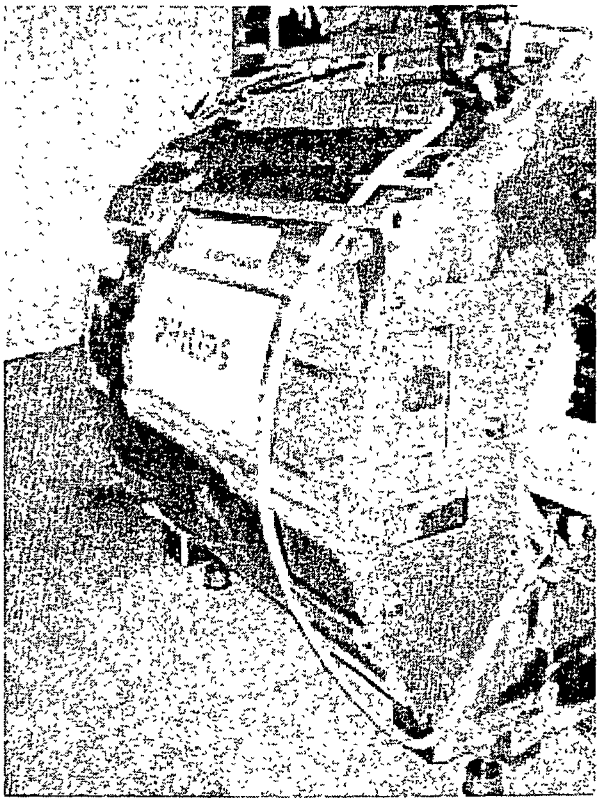


The cables coming from the Hybrid box /QBC and all cables coming from the patient support are always routed via the right side of the magnet upwards, independent of the PICU position.

Relief all cable to/from the PFEI to the magnet top-frame using tie-wraps.

Rout the cooling hose at the left side of the magnet up as shown in Figure 8.

Figure 8 – Cable routing front (left side)



6. CABLE ROUTING AT MAGNET FRONT WITH PICU MOUNTERD AT LEFT SIDE

Cable routing in this area is very critical. Be sure that no cable crossings are made. If you cross cables, magnet covers will not fit.

Figure 9 - Cable routing front left side)

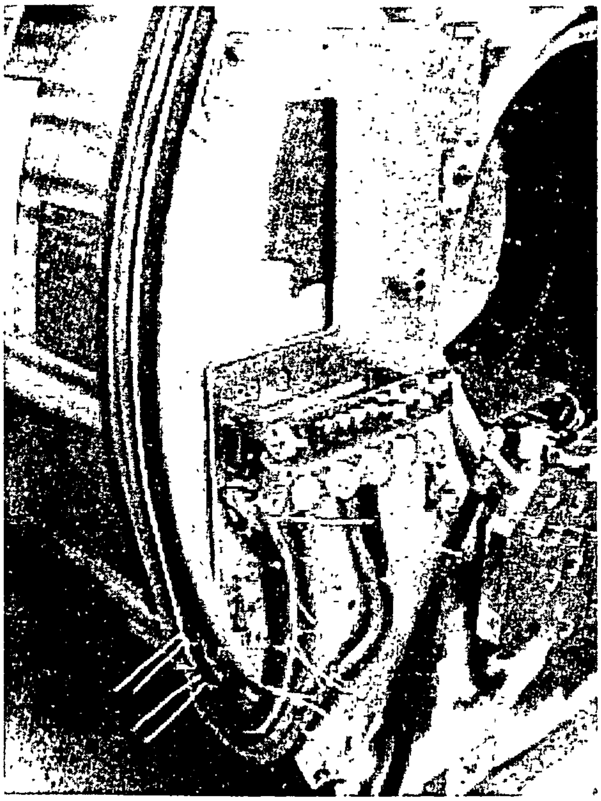
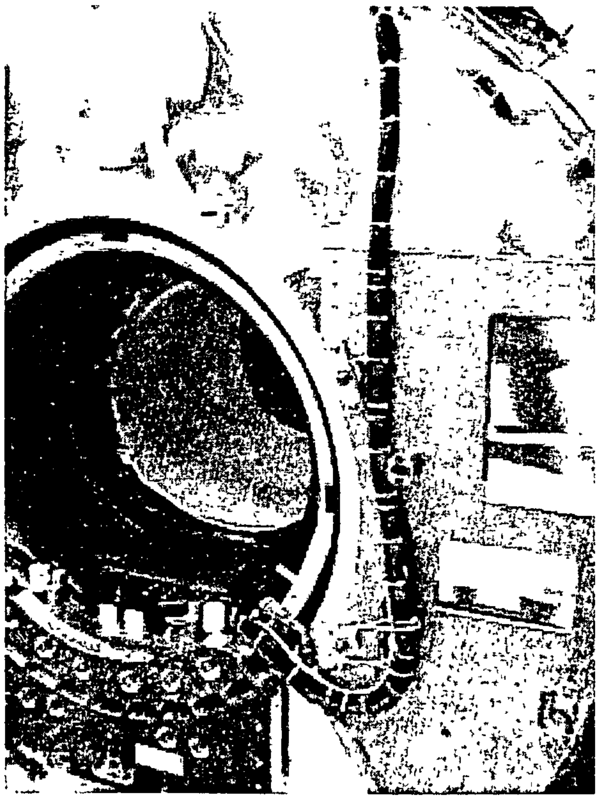


Figure 10 - Cable routing front (right side)



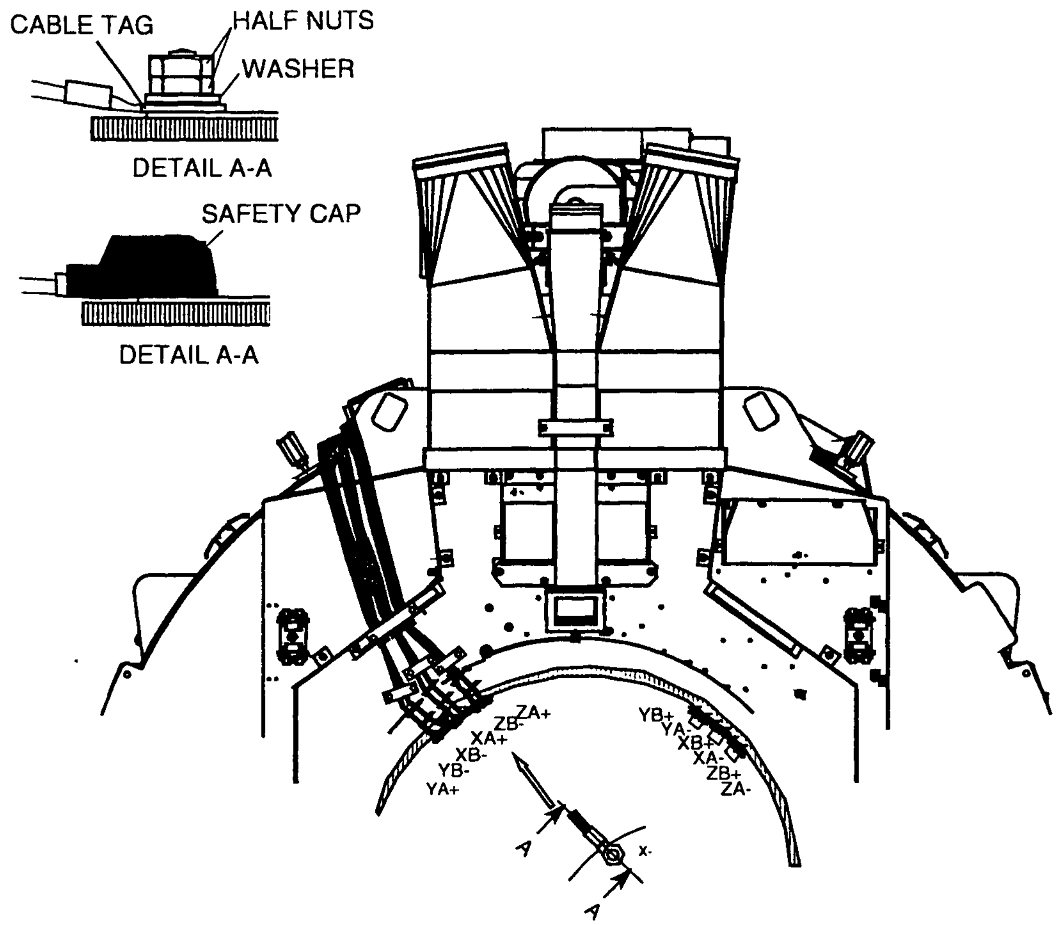
The cables coming from the Hybrid box /QBC and all cables coming from the patient support are always routed via the right side of the magnet upwards, independent of the PICU position.

Relief all cable to/from the PFEI to the magnet top-frame using tie-wraps.

7. GRADIENT CABLES AT THE MAGNET SIDE

1. The gradient cable end, which has the black sleeve (holding the cables together) is mounted to the gradient coil.

Figure 11 - Gradient cable connection



2. Shift the safety caps over the gradient cables.
3. Connect the cables to the gradient coil and tighten the nuts using a torque wrench. **Torque the first nut with 40 Nm. Apply also 40 Nm to the second nut without touching the first nut.**

NOTE

Special attention must be paid to ensure that the surfaces are clean and free from dirt and/or resin. Also the fibre glass insulation must not get caught under the cable

Also note that an incorrect connections as well as moving gradient cables near the magnet can cause image artifacts due to spikes. Apply sufficient tie-wraps to prevent cables from moving.

WARNING

If the insulation needs trimming use a small sharp scalpel. You must wear safety goggles in case the blade breaks.

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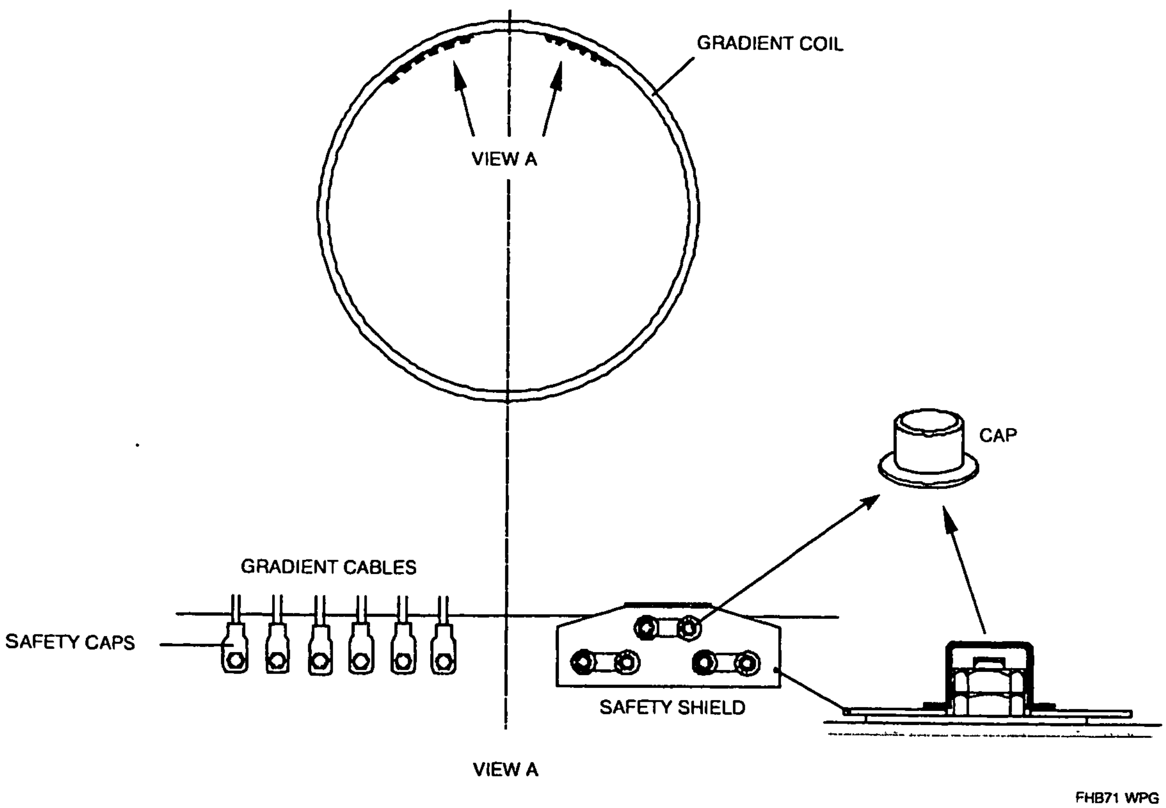
4. Shift the safety caps over the nuts.
5. Secure the cables to the ringframe with the brackets and apply tie-wraps to the gradient cables between gradient coil and the brackets.
6. Mount the safety shield over the gradient connections on the right upper side of the gradient coil.

Figure 12 - Install safety shield



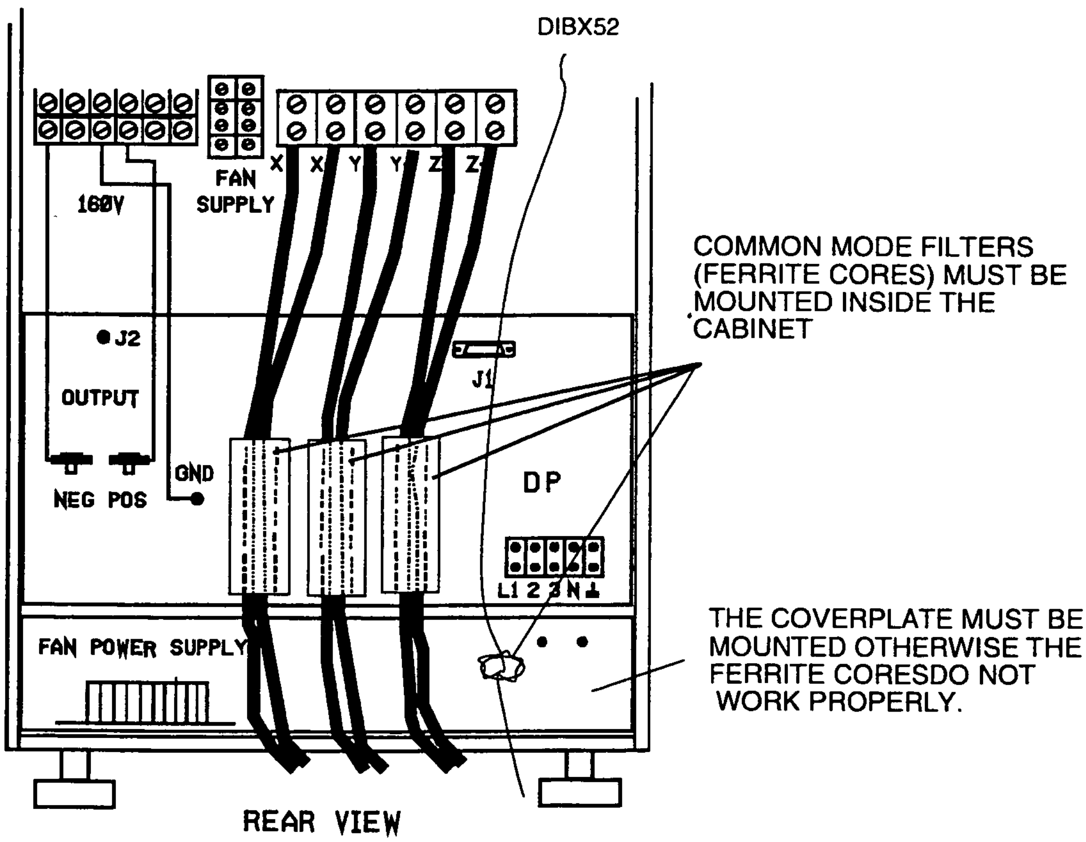
8. MOUNT COMMON MODE FILTERS INTO THE GRADIENT AMPLIFIER

Harmonic switching 81 kHz frequencies of the Copley 234 gradient amplifier are filtered using common mode filters.

Installation of common mode filters inside the copley 234 gradient amplifier around the gradient cables.
Omni system with a Copley 274 does not use these filters.

For "Standard gradients systems" shift a common mode filter over each gradient cable pair X+ X-, Y+ X- and Z+ Z-.

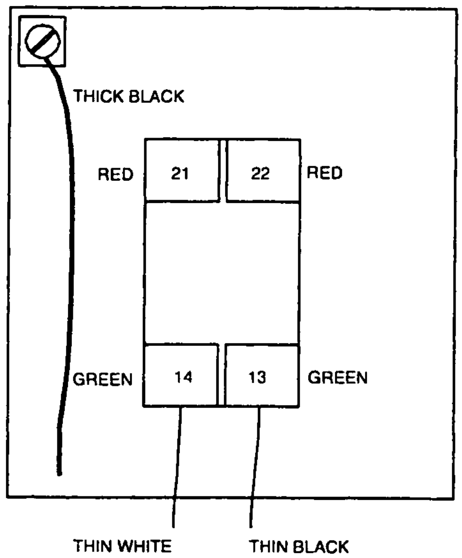
Figure 13 - Copley gradient amplifier with installed common mode filters



9. CONNECTING THE ERDU SWITCHES

The cables of the Emergency Run Down Unit switches are connected to the RMMU. The RERDU switch is "open" in the non-active state. Check this with an ohmmeter.

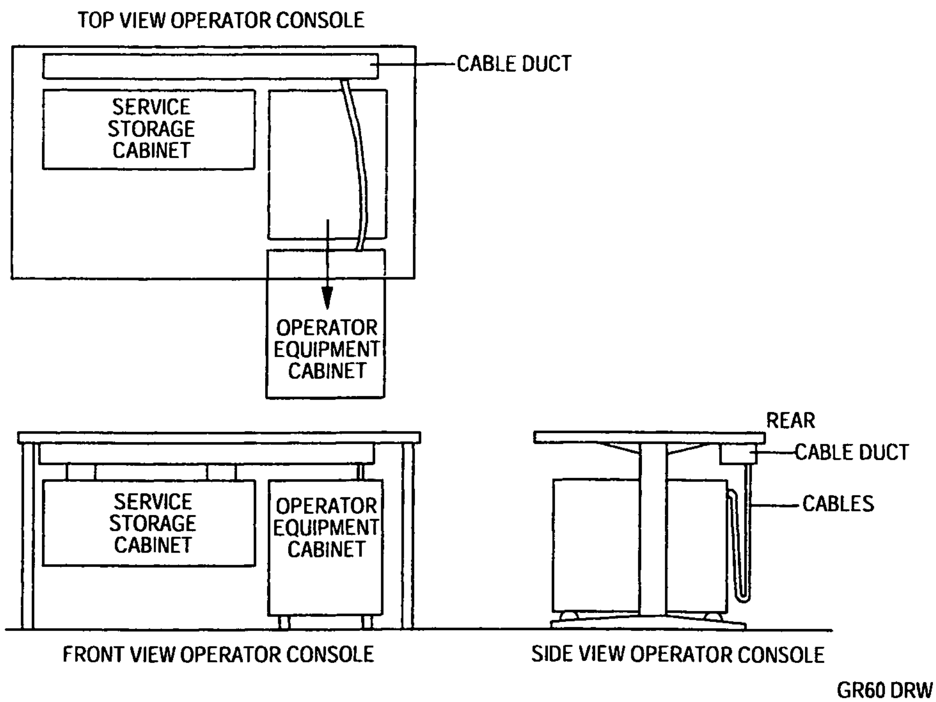
Figure 14 - Connect ERDU switch



10. CONNECTING CABLES TO THE OPERATOR CONSOLE

Normally the operator console is positioned against the wall. When the Operator Equipment Cabinet (OEC) must be accessed at the rear, the OEC must be slided forward. This is only possible if the external cabling at the rear has enough slack. For this reason the external cabling is hung in a half loop on hooks mounted on the OEC. Delivered with the console is a Mains Outlet Assy. This must be placed inside the Service Storage Cabinet on a convenient place.

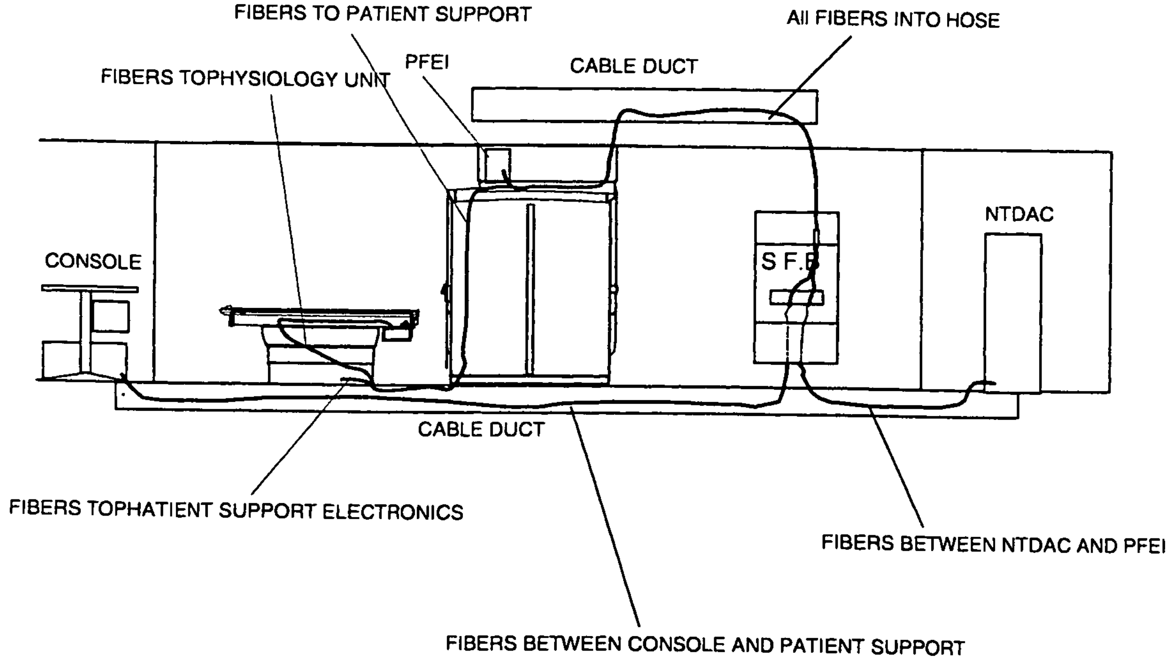
Figure 15 – Cable routing at Console



11. CONNECTING THE OPTICAL FIBRES

Install the fibers after all other cables have been installed. Route the fibres through the pipe on top of the system filter box Be careful not to bend the fiber cables too much.

Figure 16 - Routing of fibers



Section 8 Signal cables

Contents

1. GENERAL

Connect all signal cables according to the drawings:

Z5 – 2.1	Signal cable plan magnet
Z5 – 2.2	Patient support MT
Z5 – 2.4	Gradient parts (Power gradients)
Z5 – 2.5	Gradient parts (Master gradients)
Z5 – 2.6	RF parts
Z5 – 2.7	Console

Due to the fact that there is limited space between magnet covers and magnet, cables must be mounted with care. If you do not install the cables "in a nice manner" as indicated in this section, you will face problems installing the magnet covers.

Mount the optical fibers after all other cables have been installed (see chapter "Connecting the optical fibers").

CAUTION

Optic fibres are very delicate. Be careful when working with optical fibres.

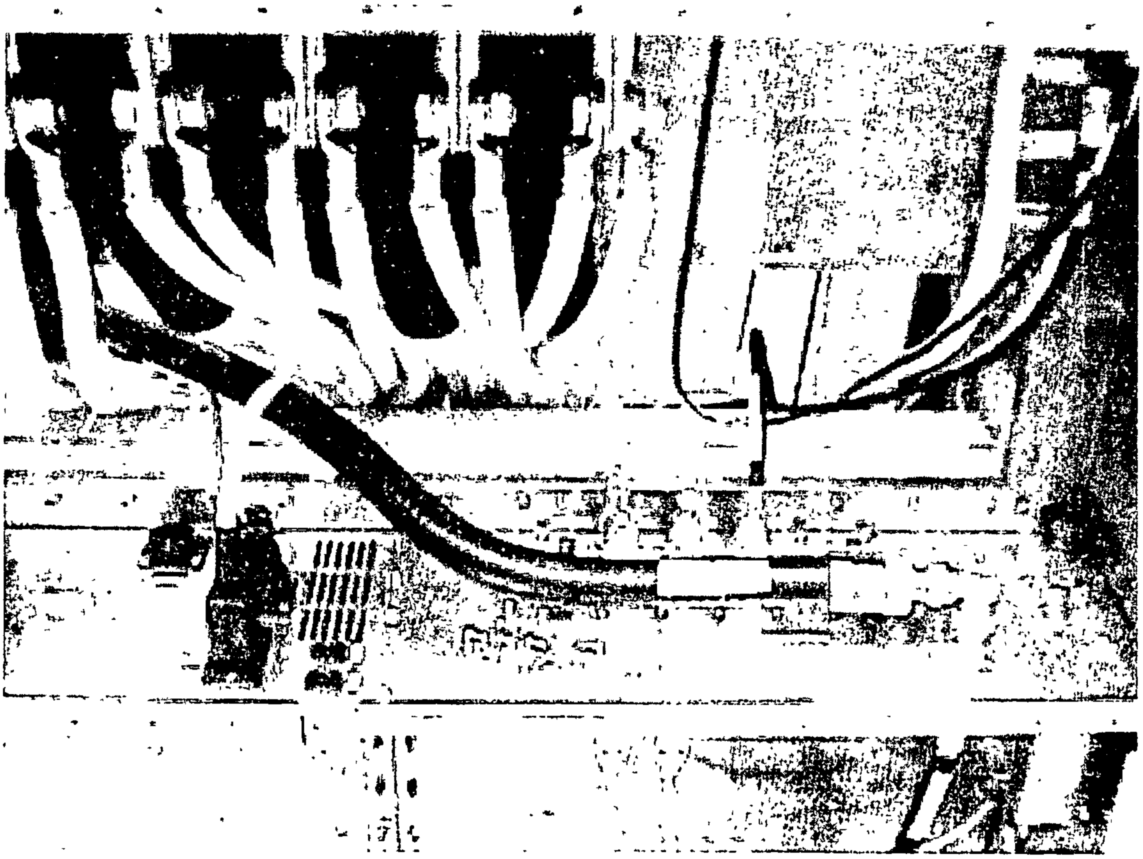
NOTE

Cables that are not connected and cables that can move against other cables can cause electrical discharge (spikes) with image artefacts as a result. Root cables properly in ducts and in the hooks, present around the magnet.

2. INSTALLATION OF THE RF HIGH POWER CABLE

Intera 1.0T or Intera 1.5T system with Power or Master use a thick high power RF power between System Filter Box and RF amplifier.

Figure 1- Root thick RF power cable



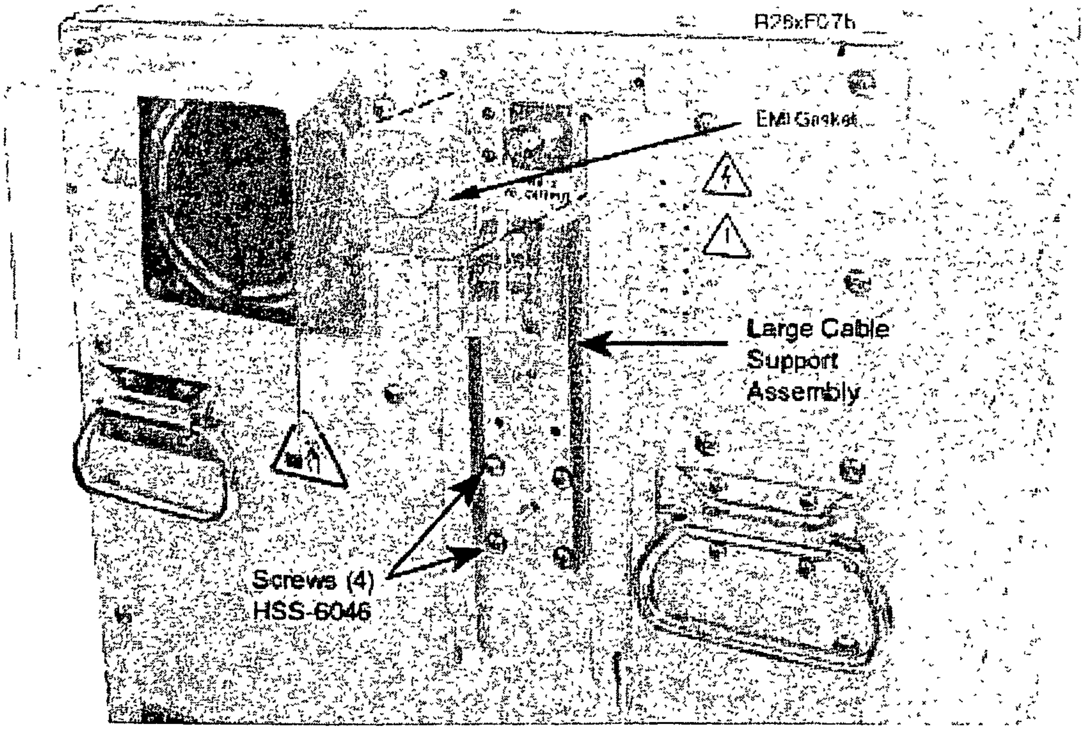
The high power cable between power amplifier and system filter box is needed in case 15 kW power is used. Note that the minimum bending radius of this stiff cable is 300 mm. The power cable has connectors on both sides. You may want to shorten the power cable. For that purpose a "service set heliax cable" is available that you can order with the 12 NC number 4522 131 54431. A connector, tools and instructions are included in this set.

In the examination room a thin RF cable type RG217 is used.

Use the delivered L piece at the RF power amplifier output connector.

If you have an S23/S24 amplifier the RF-Output is on the backside of the amplifier.

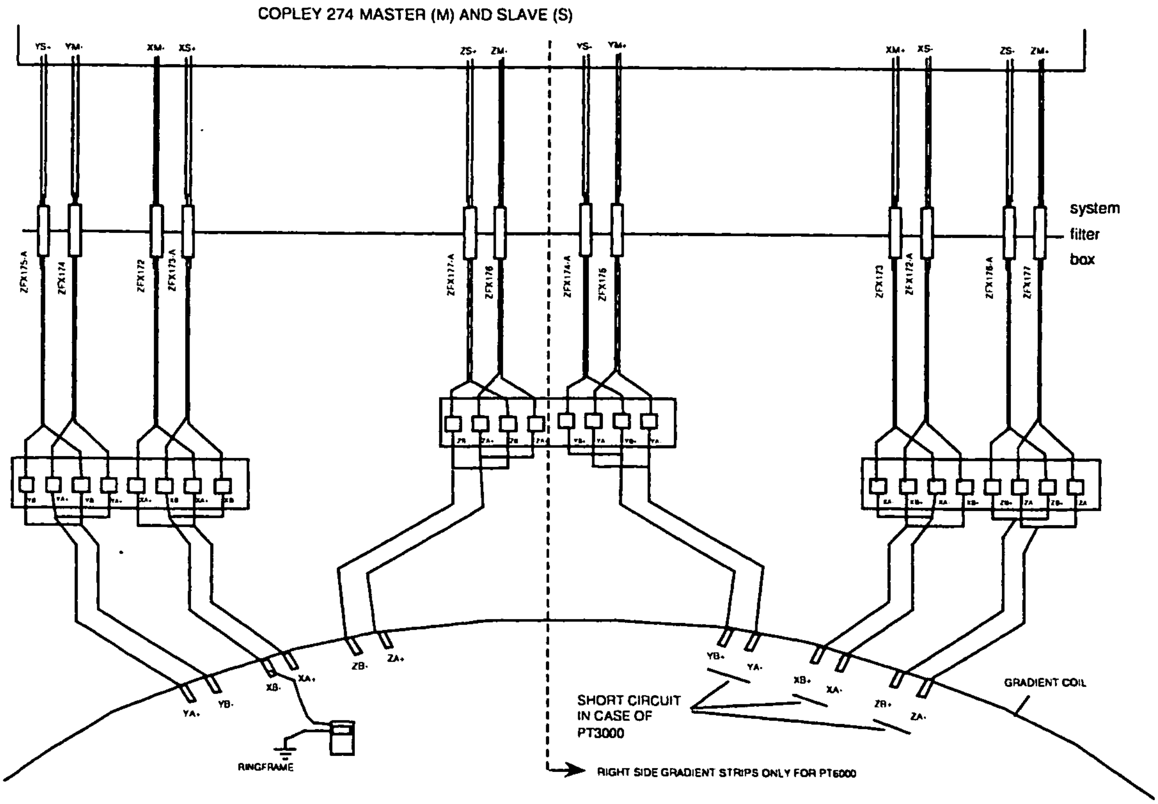
Figure 2 - showing RF output connector and cable support assembly



Gradient cables Power and Master gradients

For the Power gradients, 12 gradient cables must be installed in the examination room and in the technical room. For the Master gradients, 24 gradient cables must be installed in both the technical room and examination room. Because of this big amount of gradient cables the cable length has to be adapted. Adapting the cable length is also needed to reduce cross-stalk between these cables.

Figure 3 – Gradient cable overview



Mount the filters into the System filter box

For Power gradients, six gradient filters are factory mounted into the S.F.B. For Gyroscan Intra Master six extra gradient filters must be mounted on site. A special spanner is supplied with the filters. Also a small allen key is needed. Install the filters using drawing Z3-2.2. of the drawing manual

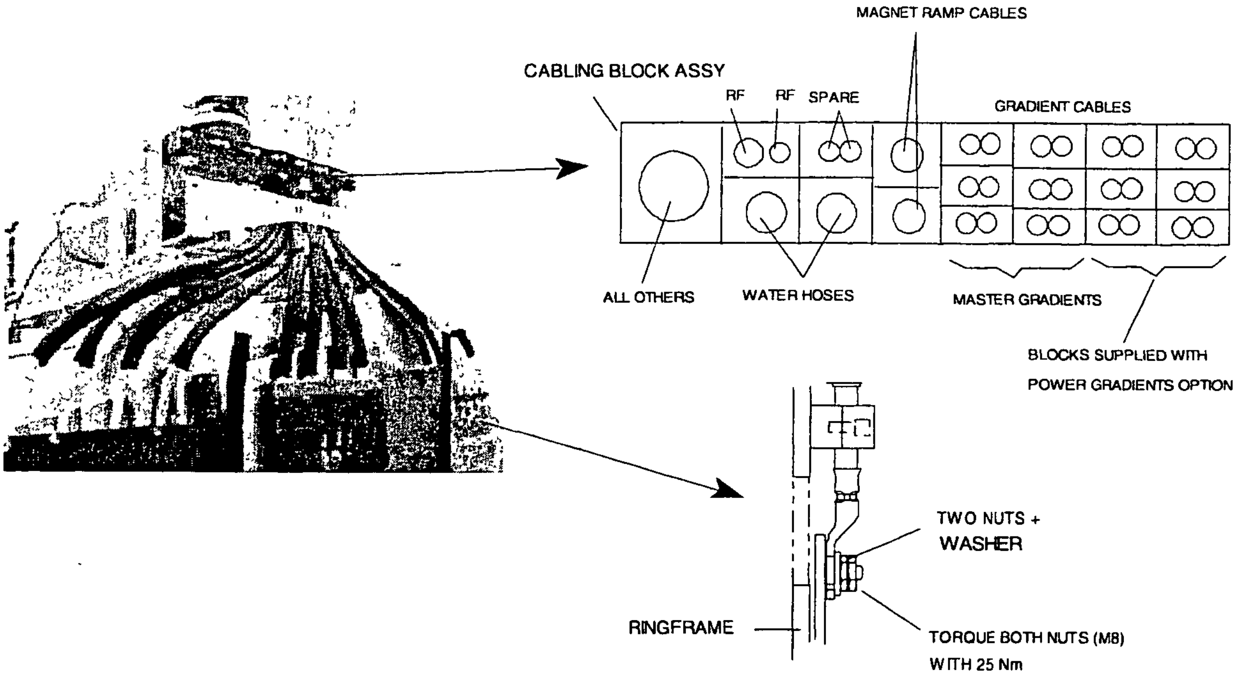
Gradient cables at rear of the magnet

Route the gradient cables through the cable relief brackets on the ringframe and the cable relief bar. The cables have to go up through the cable block at the rear side of the magnet.

The gradient cable end, which has black sleeves (holding two gradient cables together) is mounted to the Gradient strips, and the other (no cable eye) end is mounted to the system filter box.

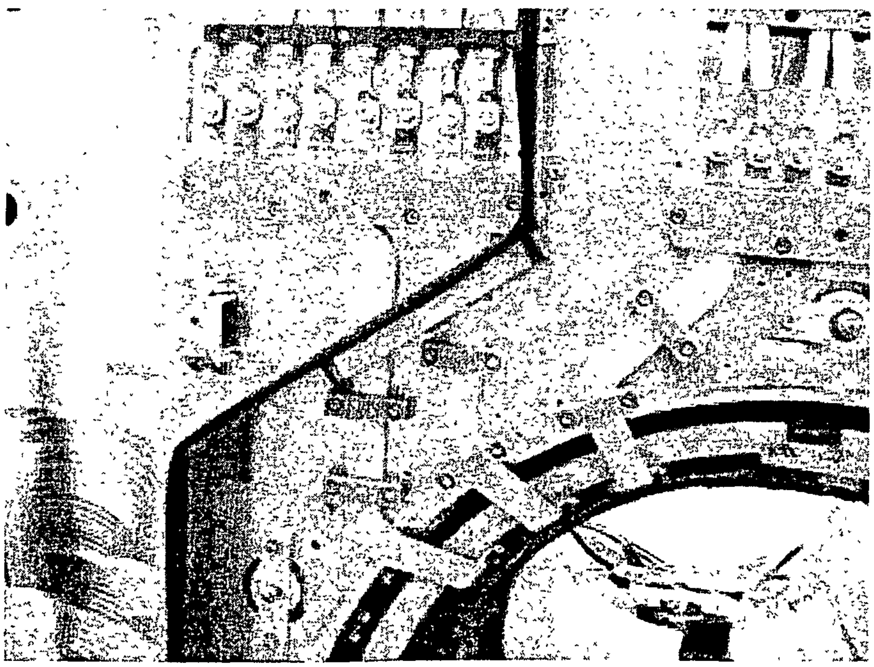
Torque the nuts (gradient cables to gradient strips) with **25 Nm**.

Figure 4 – Routing for gradient cables



MAGNET REAR

Figure 5 – Install gradient cables to gradient strips



For safety reasons, mount the three gradient strip covers.

Figure 6 - Cover left

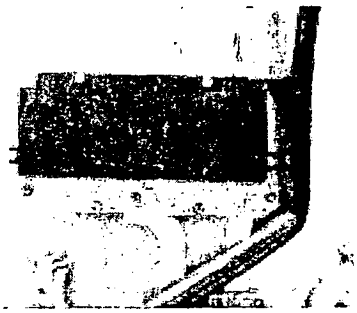


Figure 7- Middle cover

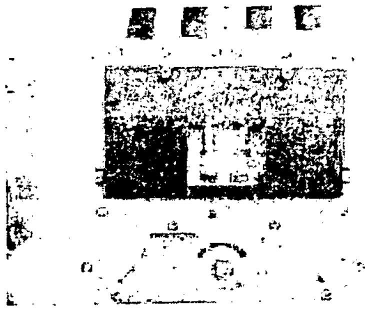
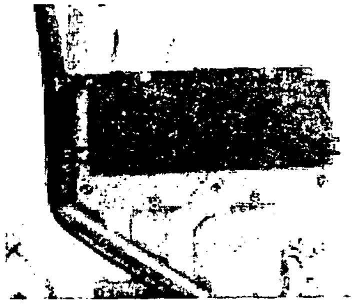


Figure 8 – Right cover



Gradient cables in System Filter Box

The cables must be shortened at the system filter box. An AMP clamp toolkit is available (see tool list in section 1). Note that each cable has a number of labels attached. When you cut a cable, be sure that you still have at least one label left.

CAUTION

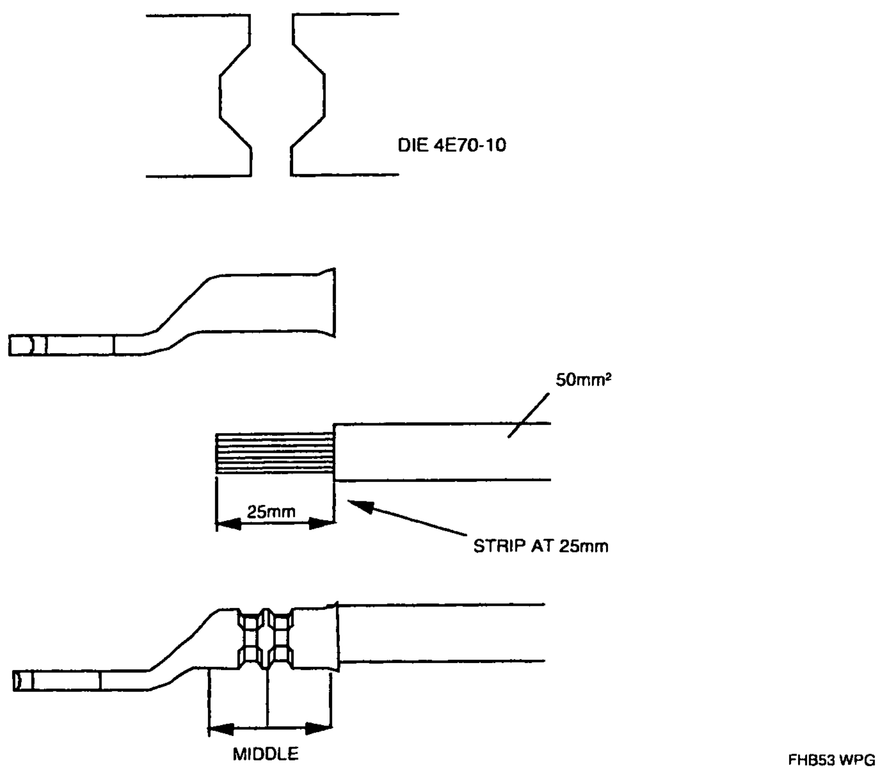
Moving gradient cables may cause image artefacts due to spikes. Therefor take the following measures:

1. *Root the gradient cables trough a separate cable duct as described in the Planned Reference Book.*
2. *Attach the gradient cables together using tie-wraps. Apply tie-wraps to the gradient cables starting at the gradient strips and stop at the position where the cables are at least 1 meter away from the top covers.*

Cut all cables with the special cutter supplied in the tool set.

Remove the cable insulation over 25 mm using the supplied stripping tool. Mount the eye tag to the cable with the "AMP clamp-tool" (see tool list in section 1). Use a "DIE 4E70-10", that is supplied with the "AMP clamp tool".

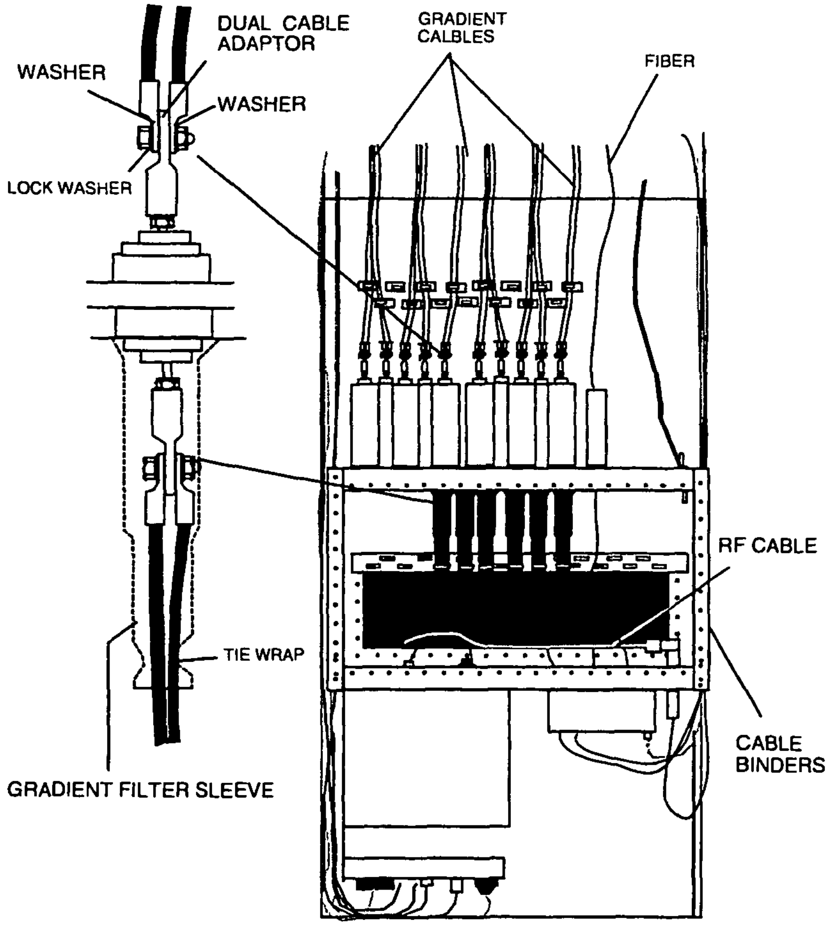
Figure 9 – Apply eye tag



Gradient filter sleeves are needed to prevent service engineers from touching the gradient connections by sticking their hand inside the system filter box from the outside of the examination (RF) room.

1) Slide the gradient filter sleeves over the gradient cables that leave the examination room (6x).
2) Connect the gradient cables to the dual cable adapters.
3) Slide the gradient filters sleeves all the way up and secure them with a tie-wrap (6x).

Figure 10 - System Filter Box



CAUTION

Take special care not to damage the gradient filters inside the system filter box. They are made of ceramic material and cannot withstand large forces; use two ring spanners 19mm to tighten the cable connection and use them in such a way that no force is applied to the filter.

Technical room

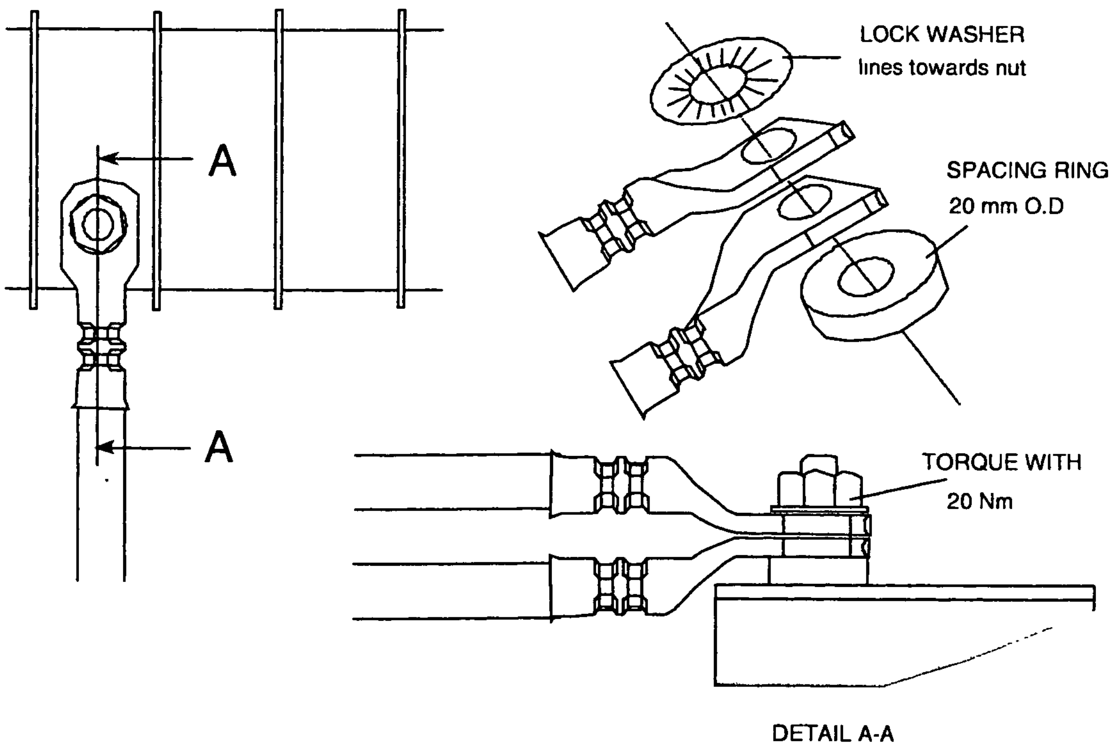
Also these gradient cables have cable tags installed at one end only. This end must be installed to the gradient amplifiers The cables must be shortened at the system filter box.

Underneath each cable tag, a spacing ring must be mounted. Without this bush, the cable tag will not fit properly into the connection terminals of the Copley amplifiers.

Note that the lock washer must be positioned with the lines towards the nut.

For the Master gradients, two cables must be mounted to one connection terminal

Figure 11 - Gradient cable connection to Copley 274



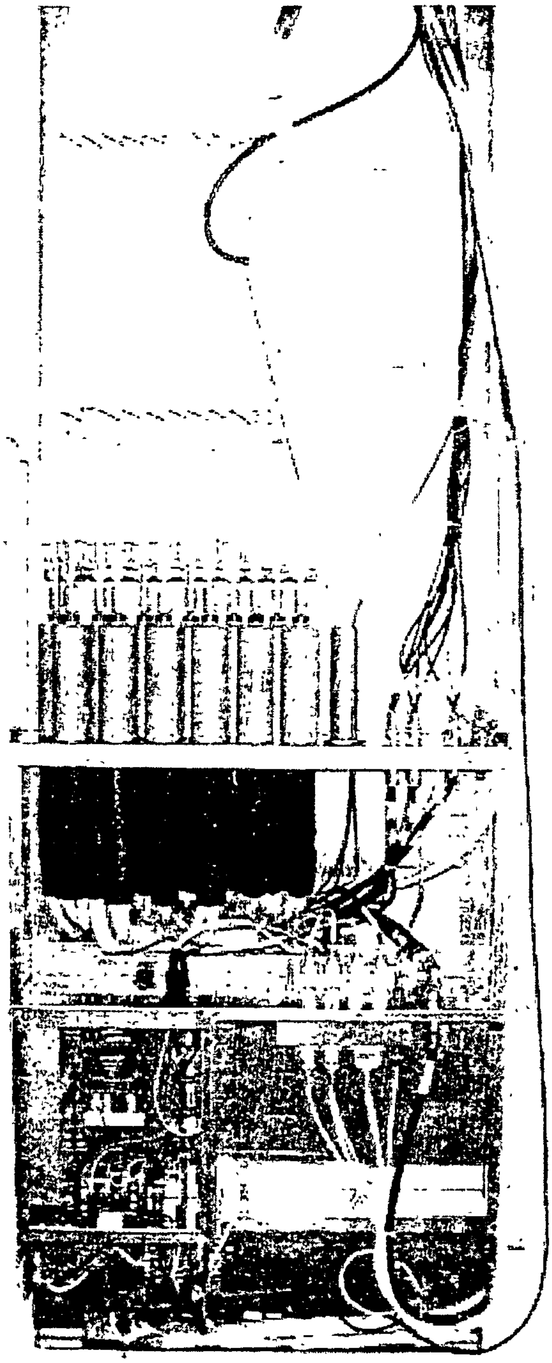
Only use a Torque wrench with a short extension and a 17-mm socket. Tighten the nuts to 20 Nm. (Even when the terminal covers are labeled with a lower torque value.)
Never use an open-ended wrench or boxed-ended wrench. Never remove the terminal block partitions while tightening the nuts. Only apply force perpendicular to the terminal block and make sure the terminal block is not skewed from perpendicular in the bracket when being tightened.

CAUTION

Do not use spacing rings other than the supplied ones. For the quality of the connection it is important to use the correct material. The connection can overheat if improper material is used.

Cables routing in the System Filter box.

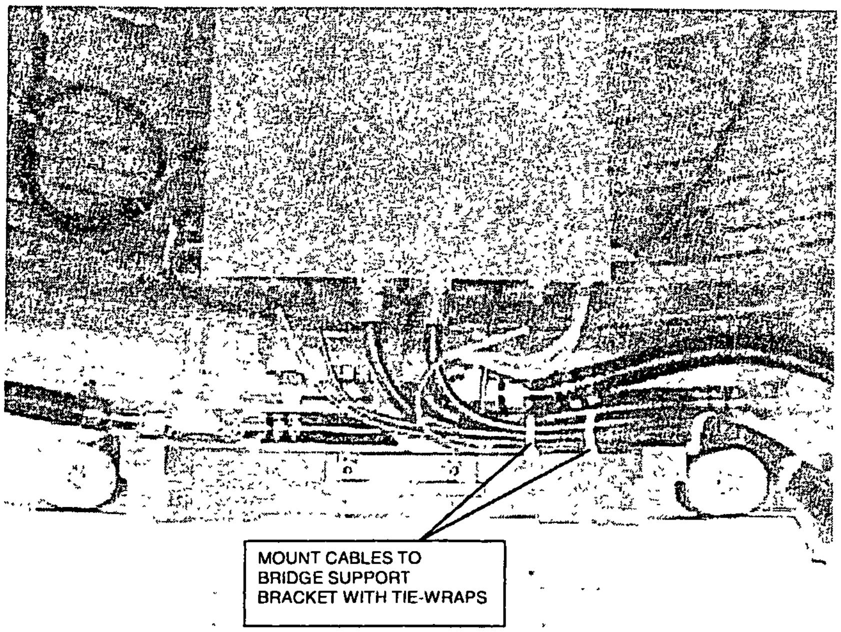
Figure 12 – Cable routing System Filter Box



3. HYBRID BOX / QBC CABLING

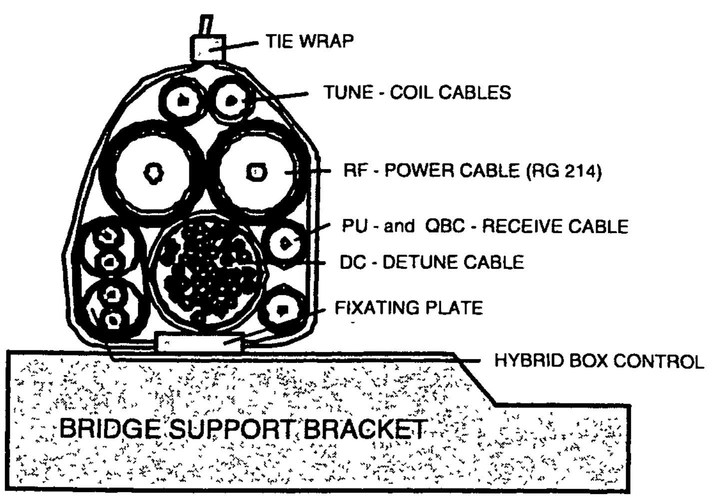
Cables from the hybrid box must always be routed via the right side of the magnet going up Attach these cables with tie-wraps to the bridge support bracket.

Figure 13 - Hybrid box cabling



The Hybrid box / QBC cables must always remain bundled as shown in the next figure.

Figure 14 – QBC/Hybrid cables



4. CABLE ROUTING AT MAGNET FRONT WITH PICU MOUNTERD AT RIGHT SIDE

Cable routing in this area is very critical. Be sure that no cable crossings are made. If you cross cables, magnet covers will not fit.

Figure 15 – Cable routing front (right side)

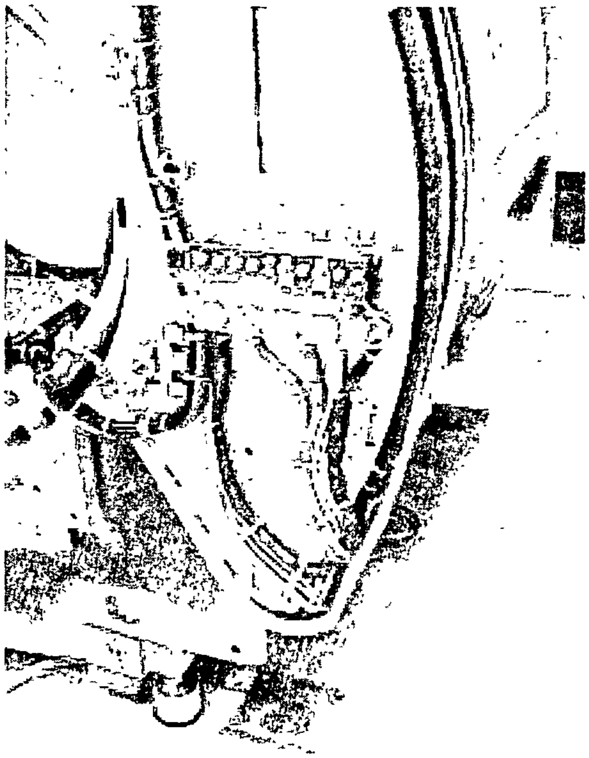
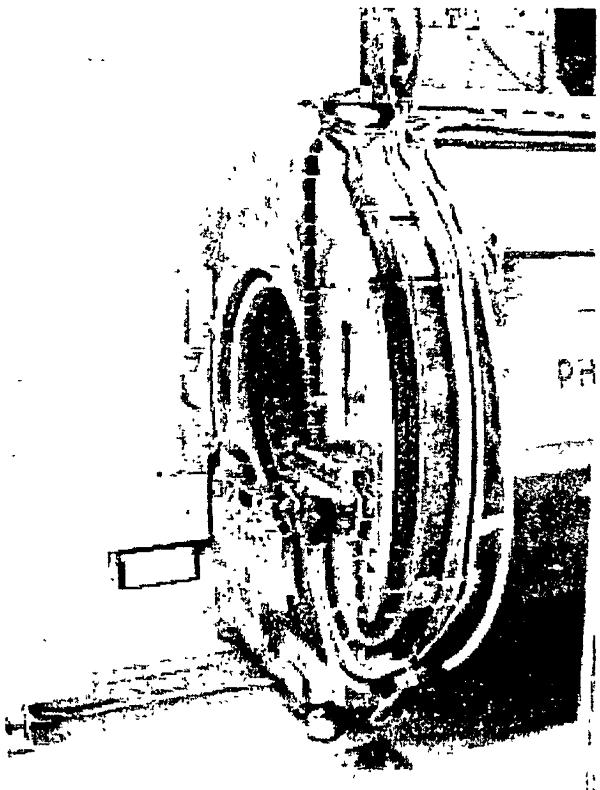


Figure 16 – Cable routing front (right side)

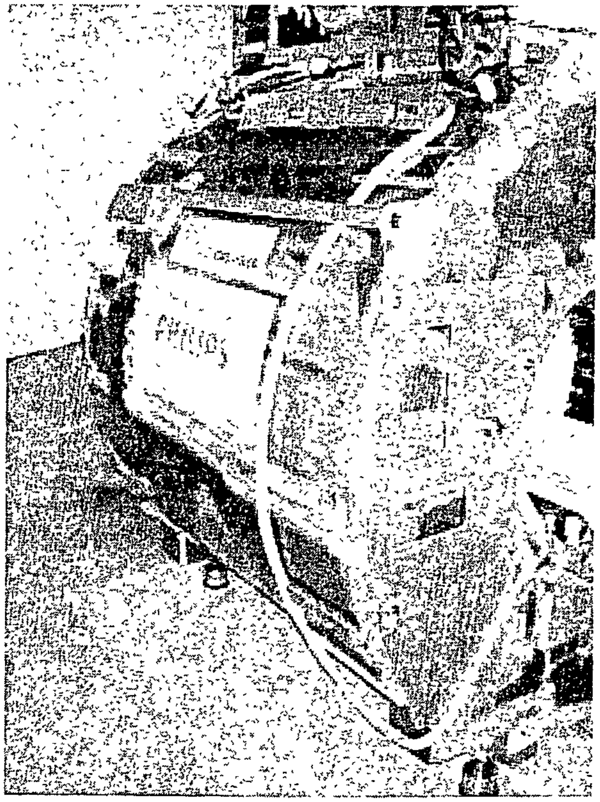


The cables coming from the Hybrid box /QBC and all cables coming from the patient support are always routed via the right side of the magnet upwards, independent of the PICU position.

Relief all cable to/from the PFEI to the magnet top-frame using tie-wraps.

Rout the cooling hose at the left side of the magnet up as shown in next figure.

Figure 17 – Cable routing front (left side)



5. CABLE ROUTING AT MAGNET FRONT WITH PICU MOUNTERD AT LEFT SIDE

Cable routing in this area is very critical. Be sure that no cable crossings are made. If you cross cables, magnet covers will not fit.

Figure 18 - Cable routing front (left side)

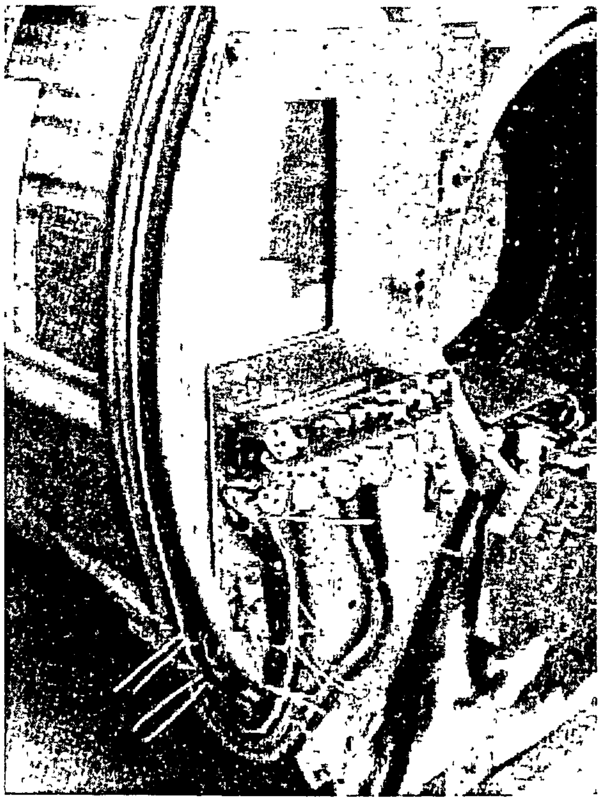
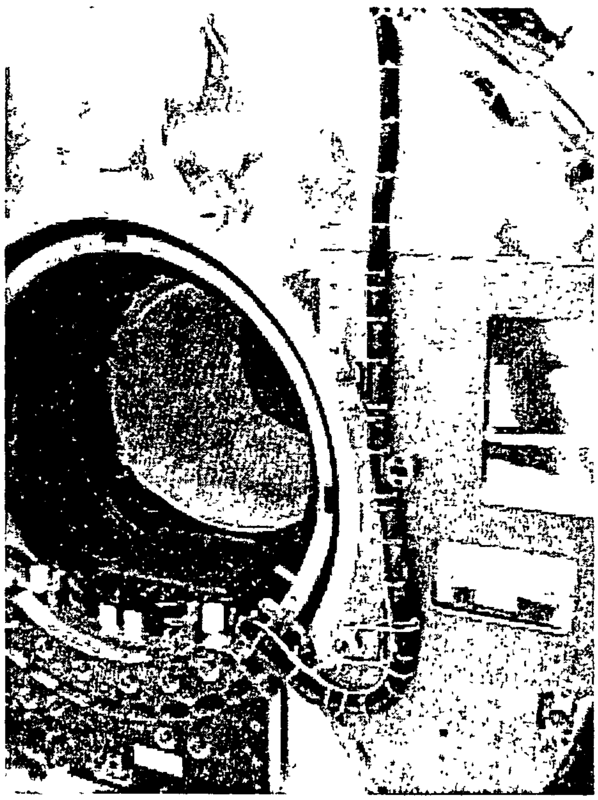


Figure 19 - Cable routing front (right side)



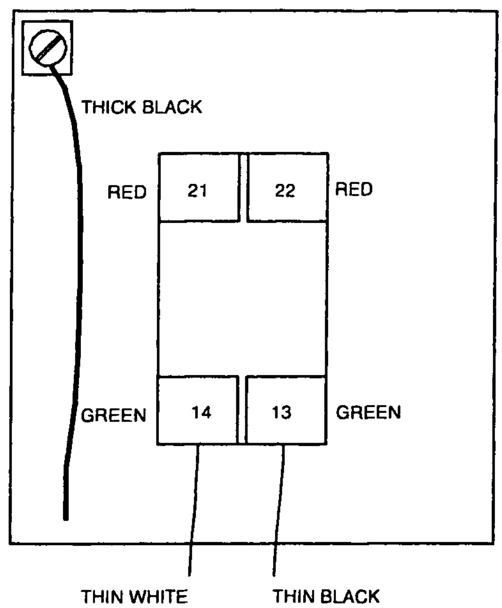
The cables coming from the Hybrid box /QBC and all cables coming from the patient support are always routed via the right side of the magnet upwards, independent of the PICU position.

Relief all cable to/from the PFEI to the magnet top-frame using tie-wraps.

6. CONNECTING THE ERDU SWITCHES

The cables of the Emergency Run Down Unit switches are connected to the RMMU. The RERDU switch is "open" in the non-active state. Check this with an ohmmeter.

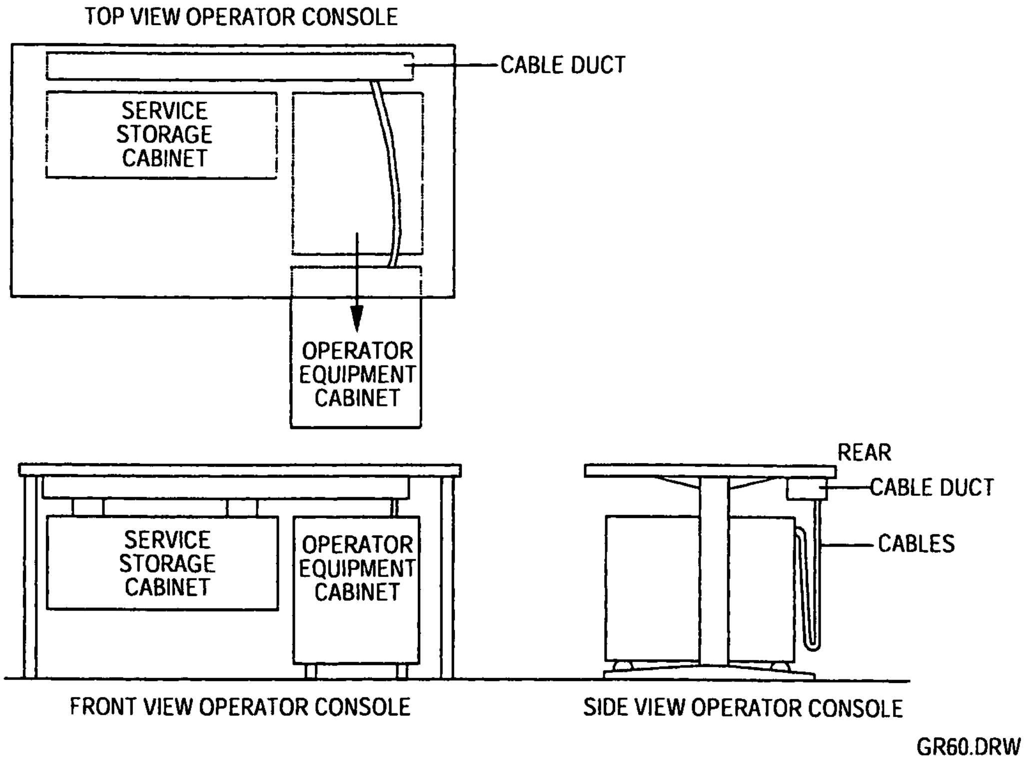
Figure 20 - Connect ERDU switch



7. CONNECTING CABLES TO THE OPERATOR CONSOLE

Normally the operator console is positioned against the wall. When the Operator Equipment Cabinet (OEC) must be accessed at the rear, the OEC must be slided forward. This is only possible if the external cabling at the rear has enough slack. For this reason the external cabling is hung in a half loop on hooks mounted on the OEC. Delivered with the console is a Mains Outlet Assy. This must be placed inside the Service Storage Cabinet on a convenient place.

Figure 21 – Cable routing at Console



8. CONNECTING THE OPTICAL FIBERS

Install the fibers after all other cables have been installed. Route the fibers through the pipe on top of the system filter box. Be careful not to bend the fiber cables too much.

Figure 22 - Routing of fibers